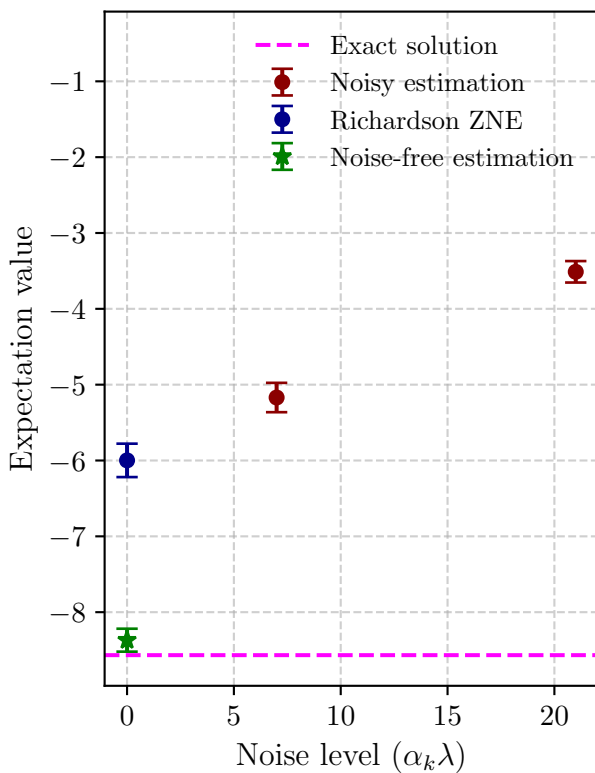
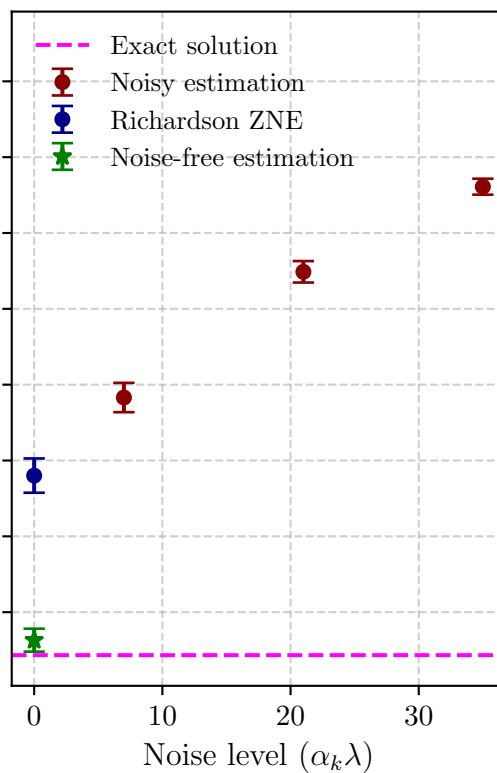


| Description | Simulation           | Mean $\pm$ Std         |
|-------------|----------------------|------------------------|
| xy-ric2     | vqe                  | -5.16950 $\pm$ 0.19350 |
| xy-ric2     | redundant (noise=7)  | -5.16950 $\pm$ 0.19350 |
| xy-ric2     | redundant (noise=21) | -3.51178 $\pm$ 0.14152 |
| xy-ric2     | zne                  | -5.99836 $\pm$ 0.22031 |
| xy-ric2     | noiseoff             | -8.36934 $\pm$ 0.15125 |
|             |                      |                        |
| xy-ric3     | vqe                  | -5.16950 $\pm$ 0.19350 |
| xy-ric3     | redundant (noise=7)  | -5.16950 $\pm$ 0.19350 |
| xy-ric3     | redundant (noise=21) | -3.51178 $\pm$ 0.14152 |
| xy-ric3     | redundant (noise=35) | -2.38931 $\pm$ 0.10534 |
| xy-ric3     | zne                  | -6.19908 $\pm$ 0.22607 |
| xy-ric3     | noiseoff             | -8.36934 $\pm$ 0.15125 |
|             |                      |                        |
| xy-ric4     | vqe                  | -5.16950 $\pm$ 0.19350 |
| xy-ric4     | redundant (noise=7)  | -5.16950 $\pm$ 0.19350 |
| xy-ric4     | redundant (noise=21) | -3.51178 $\pm$ 0.14152 |
| xy-ric4     | redundant (noise=35) | -2.38931 $\pm$ 0.10534 |
| xy-ric4     | redundant (noise=49) | -1.62821 $\pm$ 0.07951 |
| xy-ric4     | zne                  | -6.25342 $\pm$ 0.22747 |
| xy-ric4     | noiseoff             | -8.36934 $\pm$ 0.15125 |
|             |                      |                        |
| xy-ric5     | vqe                  | -5.16950 $\pm$ 0.19350 |
| xy-ric5     | redundant (noise=7)  | -5.16950 $\pm$ 0.19350 |
| xy-ric5     | redundant (noise=21) | -3.51178 $\pm$ 0.14152 |
| xy-ric5     | redundant (noise=35) | -2.38931 $\pm$ 0.10534 |
| xy-ric5     | redundant (noise=49) | -1.62821 $\pm$ 0.07951 |
| xy-ric5     | redundant (noise=63) | -1.11140 $\pm$ 0.06063 |
| xy-ric5     | zne                  | -6.26894 $\pm$ 0.22783 |
| xy-ric5     | noiseoff             | -8.36934 $\pm$ 0.15125 |
|             |                      |                        |
| xy-ric6     | vqe                  | -5.16950 $\pm$ 0.19350 |
| xy-ric6     | redundant (noise=7)  | -5.16950 $\pm$ 0.19350 |
| xy-ric6     | redundant (noise=21) | -3.51178 $\pm$ 0.14152 |
| xy-ric6     | redundant (noise=35) | -2.38931 $\pm$ 0.10534 |
| xy-ric6     | redundant (noise=49) | -1.62821 $\pm$ 0.07951 |
| xy-ric6     | redundant (noise=63) | -1.11140 $\pm$ 0.06063 |
| xy-ric6     | redundant (noise=77) | -0.75993 $\pm$ 0.04656 |
| xy-ric6     | zne                  | -6.27353 $\pm$ 0.22793 |
| xy-ric6     | noiseoff             | -8.36934 $\pm$ 0.15125 |
|             |                      |                        |
| xy-ric7     | vqe                  | -5.16950 $\pm$ 0.19350 |
| xy-ric7     | redundant (noise=7)  | -5.16950 $\pm$ 0.19350 |
| xy-ric7     | redundant (noise=21) | -3.51178 $\pm$ 0.14152 |
| xy-ric7     | redundant (noise=35) | -2.38931 $\pm$ 0.10534 |
| xy-ric7     | redundant (noise=49) | -1.62821 $\pm$ 0.07951 |
| xy-ric7     | redundant (noise=63) | -1.11140 $\pm$ 0.06063 |
| xy-ric7     | redundant (noise=77) | -0.75993 $\pm$ 0.04656 |
| xy-ric7     | redundant (noise=91) | -0.52053 $\pm$ 0.03592 |
| xy-ric7     | zne                  | -6.27492 $\pm$ 0.22796 |
| xy-ric7     | noiseoff             | -8.36934 $\pm$ 0.15125 |
|             |                      |                        |

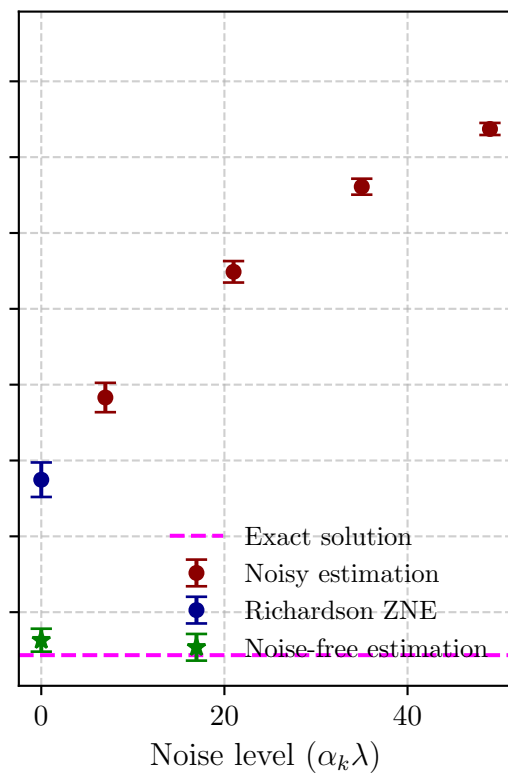
XY ansatz ric-2



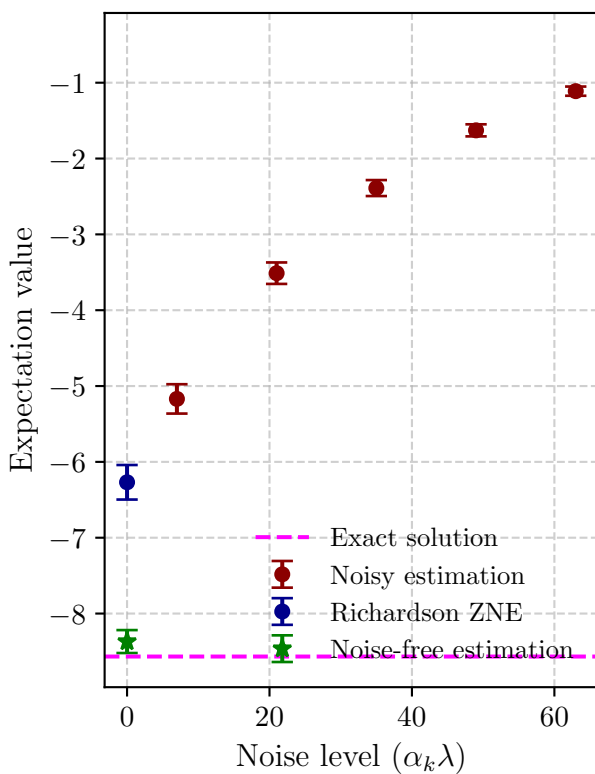
XY ansatz ric-3



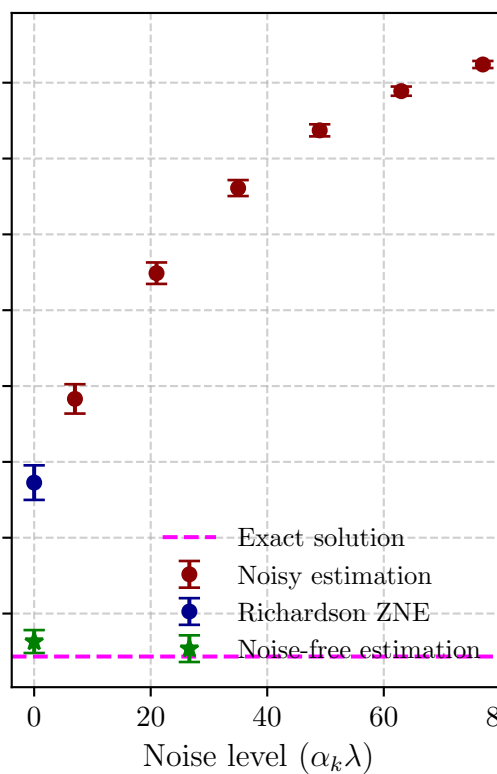
XY ansatz ric-4



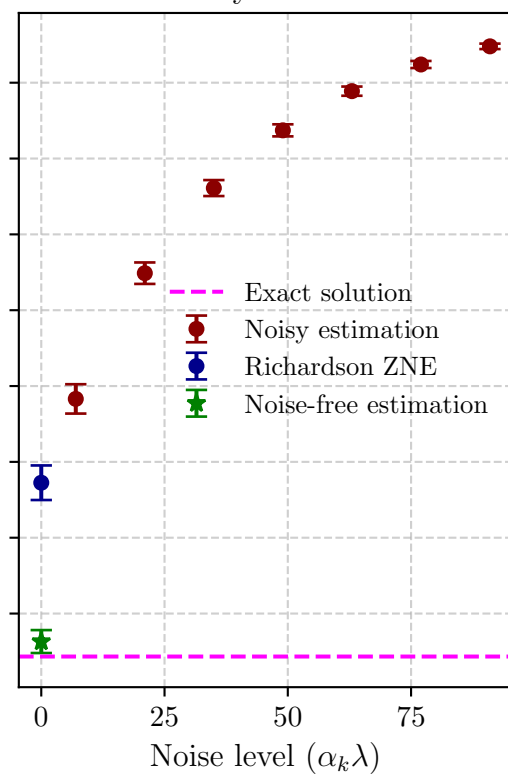
XY ansatz ric-5



XY ansatz ric-6



xy-ric7



# XY VQE CONFIGURATION

| Parameter                                | Value                               |
|------------------------------------------|-------------------------------------|
| run                                      | vqe                                 |
| nqubits                                  | 7                                   |
| state                                    | dmatrix                             |
| output.file_name_prefix                  | xy_ansatz_time_depol                |
| output.draw.status                       | False                               |
| output.draw.fig_dpi                      | 100                                 |
| output.draw.type                         | png                                 |
| observable.def                           | ising                               |
| observable.coefficients.cn               | [0.5, 0.5, 0.5, 0.5, 0.5, 0.5]      |
| observable.coefficients.bn               | [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0] |
| observable.coefficients.r                | 1                                   |
| ansatz.layer                             | 30                                  |
| ansatz.gateset                           | 1                                   |
| ansatz.ugate.type                        | xy-iss                              |
| ansatz.ugate.coefficients.cn             | [0.5, 0.5, 0.5, 0.5, 0.5, 0.5]      |
| ansatz.ugate.coefficients.bn             | [0, 0, 0, 0, 0, 0, 0]               |
| ansatz.ugate.coefficients.r              | 0                                   |
| ansatz.ugate.time.min                    | 0.0                                 |
| ansatz.ugate.time.max                    | 100.0                               |
| vqe.iteration                            | 10                                  |
| vqe.optimization.status                  | True                                |
| vqe.optimization.algorithm               | SLSQP                               |
| vqe.optimization.constraint              | True                                |
| init_param.value                         | random                              |
| noise_profile.status                     | True                                |
| noise_profile.type                       | time-depol                          |
| noise_profile.noise_prob                 | [0, 0, 0.001, 0.0]                  |
| noise_profile.noise_on_init_param.status | False                               |
| noise_profile.noise_on_init_param.value  | 0                                   |
| redundant.identity_factors               | [[0, 0, 0, 0], [1, 1, 1, 1]]        |
| zne.method                               | richardson                          |
| zne.degree                               | 1                                   |
| zne.sampling                             | default                             |
| zne.data_points                          |                                     |